

Yifan Ren

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Education

PhD in Biomedical Data Science <i>University of Wisconsin - Madison</i>	Sept 2025 - Present <i>Madison, WI</i>
Master of Science in Statistics <i>University of Wisconsin - Madison</i>	Sept 2022 - May 2024 <i>Madison, WI</i>
Bachelor of Science in Applied Math and Statistics <i>Stony Brook University</i>	Aug 2018 - Jan 2022 <i>Stony Brook, NY</i>

Work Experience

Morgridge Institute for Research <i>Machine Learning Researcher</i>	Aug 2024 - Aug 2025 <i>Madison, WI, US</i>
- Developed <i>mnDINO</i>, a vision transformer-based model for robust micronuclei segmentation in fluorescence micrographs. - Improved segmentation precision by 15% and recall by 6% over prior state-of-the-art methods. - Validated generalization across diverse unseen cell lines and microscope configurations.	

Research Experience

Graduate Researcher <i>Morgridge Institute for Research</i>	Nov 2025 - Present <i>Advisor: Juan Caicedo</i>
- Leveraged conditional flow matching to simulate cell morphological perturbations in latent space. - Designed biologically meaningful evaluation framework using KNN with Sphering normalization.	
Graduate Researcher <i>Morgridge Institute for Research</i>	July 2025 - April 2026 <i>Advisor: Juan Caicedo & Kenneth Poss</i>
- Built end-to-end pipeline to analyze Hb-egf regulation on epicardial cell dynamics in zebrafish. - Integrated CellPose-SAM and Trackastra for cell segmentation and cross-frame tracking on high-resolution fluorescence time-lapse data. - Developed color quantization algorithm to classify epicardial cell populations and compute per-cell spatial statistics relative to heart centroid. - Manuscript is under review, submitted to journal.	

Publications

- Yifan Ren**, Louise Morlot, J. Owen Andrews, Emil Peter Thrane Hertz, Niels Mailand, Juan C. Caicedo. “mnDINO: Accurate and robust segmentation of micronuclei with vision transformer networks” - Under Review.

Invited Talks

- Speaker:** Discovery Seminar Series. Morgridge Institute for Research. Madison, WI. Nov 2024

Honors & Awards

- School of Medicine and Public Health Scholarship, University of Wisconsin - Madison

Skills

- ML/DL:** PyTorch, scikit-learn, Hugging Face, OpenCV, CellPose, Trackastra
- Languages:** Python, R, SQL, Bash
- Infrastructure:** Docker, CHTC (HTCondor), SLURM, Git, Linux